Lecture 18, DAA   Mar 25

	1	2	3	4
P	10	10	12	18
W	2	4	6	9

$B = 15$

$K(i, y)$: best soln for $\{1, 2, \dots i\}$ and capacity $y \leq B$

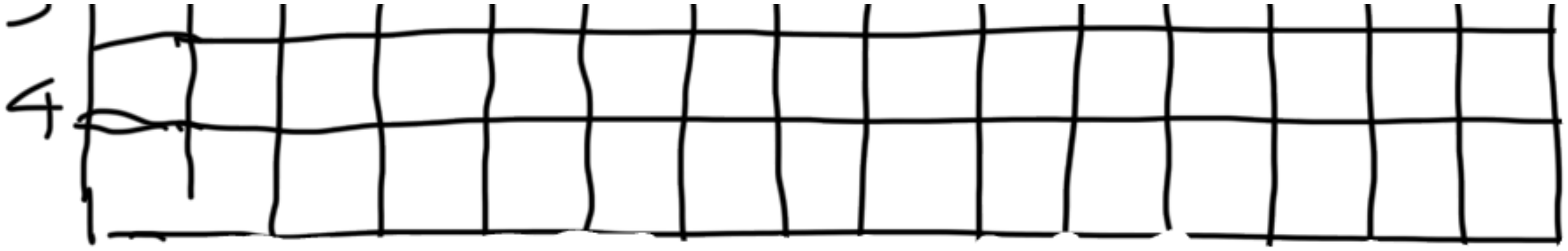
$$K(i, y) = \max\{K(i-1, y), K(i-1, y - w_i) + p_i\}$$

$$K(1, j) = p_1 \text{ if } w_1 \leq j, \ 0 \text{ otherwise}$$

$$K(i, j) = -\infty \text{ if } j < 0 \text{ (infeasible)}$$

i \ y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	0	10	10	10	10	10	10	10	10	10	10	10	10	10	10
2															
3															

Size of table
$n \times B$
Time to fill



each entry $= O(1)$

$\Rightarrow O(n \cdot B)$

$= O(n \cdot 2^{\log B})$

To do: Write an alternate DP using i and p_i (instead of w_i)

DP & DnC

1. We start by writing a recurrence for the optimal soln : Principle of Optimality

   Correctness is established

2. The recurrence for optimal defines a table of some dimensions and it can be filled up efficiently to yield the final opt solution (one cell of the table)

3. The current cell can be filled up from previously computed cells.

Longest increasing Monotonic subsequence

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
	13	5	8	12	9	14	15	2	20
$g(i)$	1	1	2, 2	3, 3	3 3	4, 5	5 6	1	6 7

1 : refers to the previous index of the longest sequence

Erdos Szekeres: Given a sequence of n arbitrary numbers, the longest increasing/decreasing sequence has length $\geq \lceil \sqrt{n} \rceil$

Let $g(i)$ denote the length of the longest increasing subsequence ending in position i and includes x_i

Intuition : about $g(i)$ - If $L(i)$ is the longest increasing sequence ending at x_i and $x_i^{prev} = x_j$ is the previous term of $L(i)$,

then $\underbrace{x_{i_1}, x_{i_2}\ x_{i_3} \dots x_j}$ $\leq x_i$

$i_1 < i_2 < i_3 < \dots$

must also be the longest incr. seq ending at x_j (otherwise contradiction)

Running time for computing $g(i) = \sum (i-1)$

So for computing all $g(i)$s $\sum_{i=1}^{n} (i-1) = O(n^2)$

Question : Can we improve this using some clever data structure?